A Dual-Model Machine Learning Framework for Pavement Rutting and Condition Classification

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Abstract: - This study introduces a dual-model framework using Long-Term Pavement Performance (LTPP) data to predict pavement rutting and classify overall condition based on common variables (ESALs, thickness, temperature, precipitation, drainage). A K-Nearest Neighbors (K-NN) regression model predicted rut depth with reasonable accuracy (Sum of Squared Errors = 205.03), particularly for moderate rutting (2–5 mm). Concurrently, a Decision Tree classification model (CART algorithm) categorized pavement condition ("Good"/"Unsatisfactory") based on International Roughness Index (IRI) thresholds, achieving 81.25% accuracy and 90.91% recall. Pavement thickness, rut depth, and temperature emerged as the most influential factors for condition classification. The Decision Tree and K-NN both showed persistent accuracy, proving particularly effective for identifying pavements in good condition. This research highlights the potential of machine learning for integrated pavement management and suggests areas for future improvement, including dataset expansion and refining variable definitions.

1. Introduction

Pavement performance is a key part of managing transportation infrastructure, helping agencies prioritize maintenance, ensure road safety, and control long-term costs. Reliable tools like rut depth measurement and the International Roughness Index (IRI) allow decision-makers to plan maintenance schedules, rehabilitation strategies, and budgeting more effectively. Rutting refers to the longitudinal depressions that form in wheel paths, mainly due to repeated traffic loads, and is influenced by factors like material type, pavement thickness, temperature, and moisture conditions. IRI, meanwhile, measures surface irregularities that affect ride comfort, fuel efficiency, and vehicle wear.

In recent years, predictive modeling has become increasingly important for understanding how different variables drive pavement deterioration. Among these, Equivalent Single Axle Loads (ESALs) consistently stand out as a major traffic-related factor, providing a standardized measure of cumulative vehicle load impacts. Early studies showed a strong relationship between ESALs and rut depth, highlighting traffic as a key driver of pavement deformation (Finn et al., 1976). Later research, including analysis of Long-Term Pavement Performance (LTPP) data, demonstrated that incorporating additional factors like pavement thickness, climate conditions, and drainage could significantly improve prediction accuracy (FHWA, 2006; Al-Khateeb et al., 2023).

Pavement thickness, especially for asphalt concrete (AC) surfaces and base layers, plays a critical role by distributing traffic loads more effectively and reducing stress on the subgrade. Thicker pavements are generally more resistant to permanent deformation, although the benefits tend to level off beyond certain thicknesses, making it important to balance durability with construction

costs (Jaafar & Uddin, 2016). Environmental factors also have a strong influence: high temperatures soften asphalt, making it more vulnerable to rutting, while freeze-thaw cycles and excess moisture can weaken both asphalt and subgrade materials. In particular, poor drainage can trap moisture, accelerating pavement deterioration (Lajnef et al., 2020).

Good drainage systems are crucial for protecting pavements against moisture damage. Roads with proper drainage consistently perform better over time by maintaining the strength of the underlying layers (FHWA, 1993; Sousa et al., 2023). However, many models still oversimplify drainage by treating it as a simple yes-or-no factor. There's a growing need for better ways to measure and represent drainage impacts in predictive models.

At the same time, the use of modern data-driven approaches to model pavement performance has expanded rapidly. Techniques that use structural, traffic, environmental, and age-related data have achieved strong success in predicting pavement conditions based on IRI. Agencies often categorize pavements with an IRI below 1.5 m/km as being in "good" condition, while higher values indicate "fair" or "poor" condition depending on severity (FHWA, 2017). Models built with these types of data consistently achieve high accuracy, reaffirming the importance of factors like ESALs, thickness, temperature, and pavement age (Ghanim et al., 2023; Gedafa et al., 2021; Mazari et al., 2021).

Still, important challenges remain. Most notably, there is a need for models that can predict both rutting and roughness using the same set of input variables, making pavement management more efficient and coordinated. Regional calibration is also lacking, particularly for areas with unique climate conditions. And despite its importance, drainage is still not captured with enough precision in many studies.

This study addresses these gaps by developing two models: a K-nearest neighbor model to predict rut depth and a classification model to predict pavement condition based on IRI thresholds. Both models use the same set of variables, ESALs, pavement thickness, temperature, precipitation, and drainage presence, to provide a comprehensive view of pavement performance. By combining traditional statistical techniques with newer data-driven methods, this approach aims to improve both interpretability and accuracy while offering a practical framework for transportation agencies to apply in future projects.

2. Objectives

The primary objective of this study is to develop data-driven predictive models for evaluating pavement performance by analyzing the impact of structural and environmental factors. Using machine learning techniques, specifically K-Nearest Neighbors (K-NN) and Decision Tree algorithms, the study aims to model both rut depth prediction and pavement condition classification. By leveraging variables such as Equivalent Single Axle Loads (ESALs), pavement thickness, temperature, precipitation, and drainage presence, the models are designed to support more informed, proactive decision-making in pavement maintenance and management.

The specific objectives of the study are to:

- Identify key factors that significantly influence pavement rutting and overall condition.
- Develop a K-NN model to predict rut depth using the identified input variables.

- Build a Decision Tree classification model to categorize pavement segments into “Good” or “Unsatisfactory” condition based on structural and environmental features.
- Evaluate the accuracy and performance of both models using appropriate statistical metrics such as sum of squared errors, accuracy, precision, recall, and F1-score.
- Interpret model outputs to understand the relative importance of each variable in influencing rutting and condition outcomes.

3. Data Collection and Methodology

The data for this study was collected from the Long-Term Pavement Performance (LTPP) database, an extensive pavement performance archive maintained by the Federal Highway Administration (FHWA). Access to the data was obtained through the InfoPave portal, which offers user-friendly access to pavement structural, environmental, traffic, and performance datasets collected from across the United States.

For this project, key variables were selected, including Equivalent Single Axle Loads (ESALs), pavement thickness, average annual temperature, annual precipitation, and drainage presence. These variables were extracted from different LTPP datasets and merged into a single database using the Strategic Highway Research Program Identifier (SHRP_ID), which uniquely identifies each pavement section. This merging process ensured that information from multiple sources corresponded accurately to the same pavement sections.

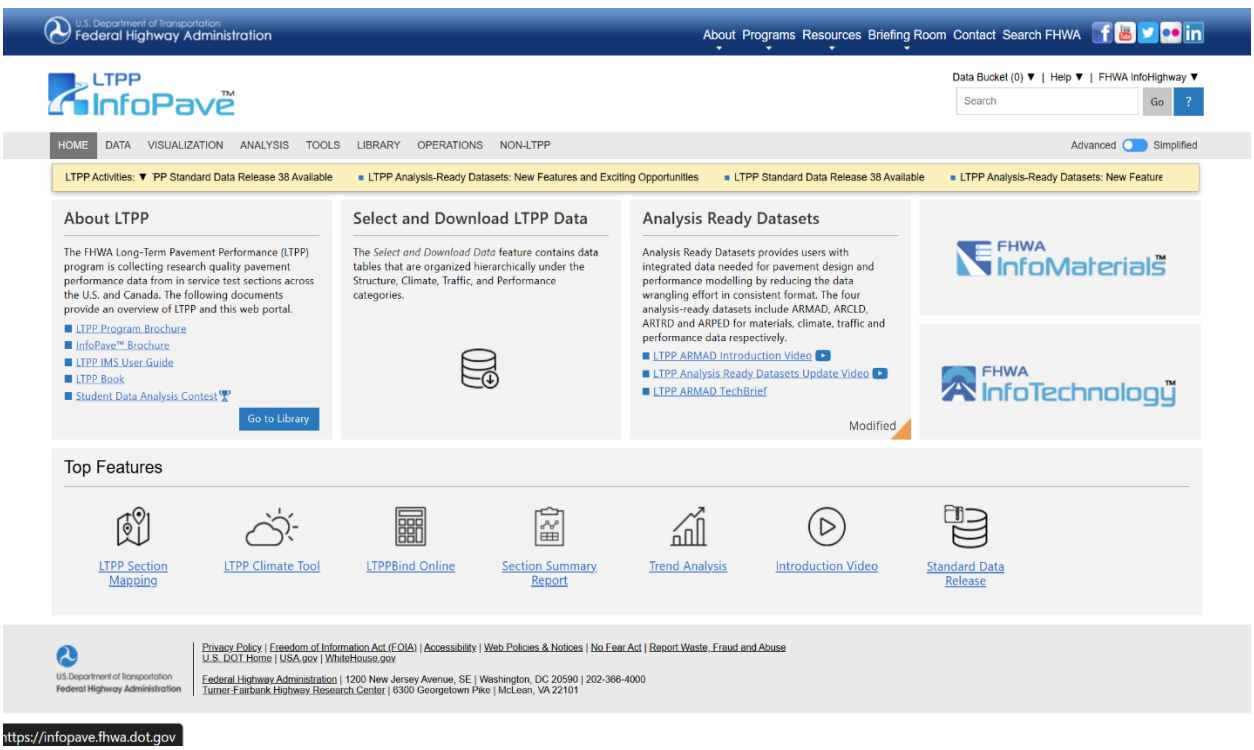

Figure 1: A snapshot of the LTPP InfoPave website

After carefully combining the datasets, the final collections included:

- 187 pavement sections used for training the predictive models.

- 45 pavement sections reserved for testing the rut depth prediction (k-NN) model.

- 48 pavement sections used for testing the pavement condition classification model.

The data merging process involved not only matching the SHRP_IDs but also performing consistency checks to verify the completeness and reliability of each record. While the LTPP "analysis-ready" datasets required minimal data cleaning, any sections lacking critical information for the selected variables were excluded from the study to maintain data integrity.

For the rut depth prediction component of the study, the K-Nearest Neighbors (K-NN) algorithm was implemented using Python within the PyCharm integrated development environment. The model was built using libraries such as scikit-learn for the machine learning algorithm and pandas for data handling. This setup enabled the development of a data-driven model capable of predicting continuous rut depth values based on pavement and environmental inputs. Figure 2 shows the PyCharm interface with the code loaded, illustrating the development environment and logic used to train and test the model.

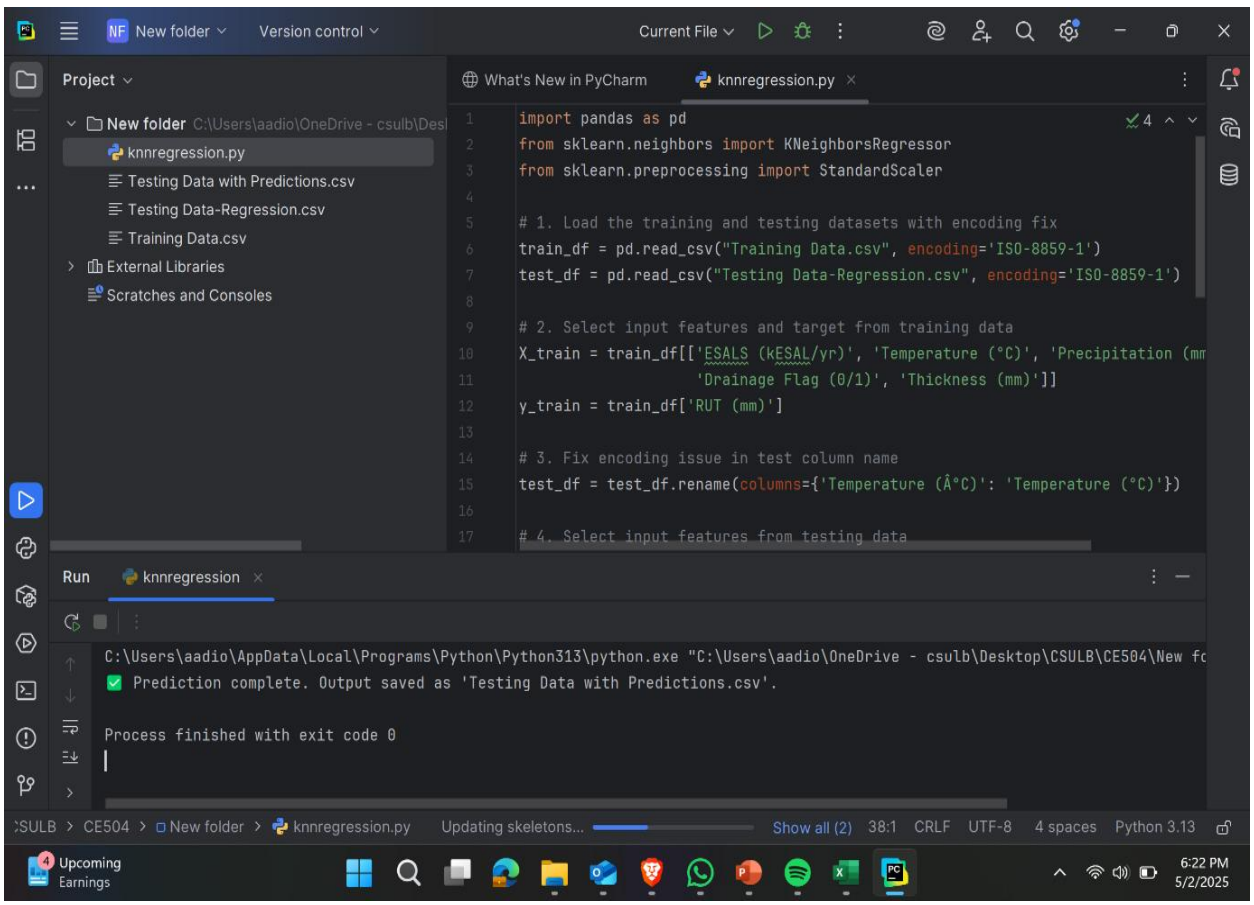

Figure 2: A snapshot of PyCharm Software running Python

For the classification of pavement condition, the study utilized the CART (Classification and Regression Tree) algorithm available in Minitab 22. Minitab's graphical interface and built-in analytical tools facilitated the construction of an interpretable decision tree, using entropy-based splitting criteria. This approach allowed for effective evaluation of variable importance and clear classification logic for labeling pavement segments as either "Good" or "Unsatisfactory." Figure 3 presents the Minitab interface used during this phase of the project.

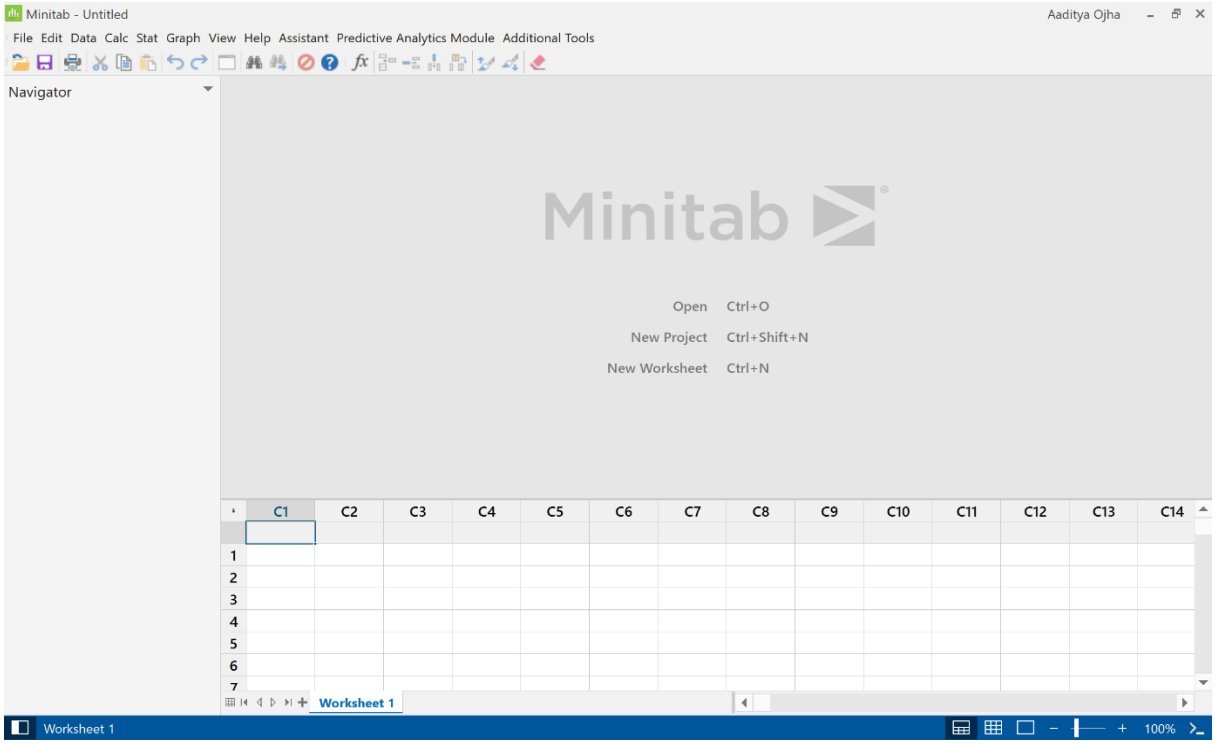

Figure 3: A snapshot of Minitab Statistical Software

4. Data Analysis

4.1 kNN Analysis

A kNN model was developed to predict pavement rut depth based on various structural and environmental factors. The independent variables selected were Equivalent Single Axle Loads (ESALs), pavement thickness, average annual temperature, annual precipitation, and the presence of drainage systems. Table 1 provides a representative sample from the training dataset to illustrate the nature of the data used. The complete training dataset, consisting of 187 pavement sections, is available in Appendix A.

Table 1: Sample of Training Dataset

State	ESALs (kESAL/yr)	Drainage Flag (0/1)	Thickness (mm)	Average Annual Temperature (°C)	Average Annual Precipitation (mm)	Rut Depth (mm)
Arizona	541	0	6.85	23.3	185.3	6
California	1291	0	7.15	17.7	179.2	9
Delaware	221	0	16.16	14.1	1089.2	2
Indiana	572	1	2.59	13.5	1563.3	8
Iowa	411	0	12.93	9.6	1107.7	2

For the K-NN model, data analysis was performed using Python to explore how key pavement and environmental variables influence rutting depth. The dataset included five predictors: ESALs, pavement thickness, temperature, precipitation, and drainage presence. Each variable was examined for consistency and relevance before modeling.

Figure 4 displays the PyCharm interface with the complete code used to conduct this analysis, showing the standardized data input, model training, and prediction steps.

```python
import pandas as pd
from sklearn.neighbors import KNeighborsRegressor
from sklearn.preprocessing import StandardScaler
# 1. Load the training and testing datasets
train_df = pd.read_csv("Training Data.csv", encoding='ISO-8859-1')
test_df = pd.read_csv("Testing Data-Regression.csv", encoding='ISO-8859-1')
# 2. Select input features
features = ['ESALS (kESAL/yr)', 'Temperature (°C)', 'Precipitation (mm)',
            'Drainage Flag (0/1)', 'Thickness (mm)']
X_train = train_df[features]
y_train = train_df['RUT (mm)']
X_test = test_df[features]
# 3. Standardize the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
# 4. Train kNN model
knn = KNeighborsRegressor(n_neighbors=3)
knn.fit(X_train_scaled, y_train)
# 5. Predict and save results
test_df['Predicted_RUT (mm)'] = knn.predict(X_test_scaled)
test_df.to_csv("Testing Data with Predictions.csv", index=False)
print("✅ Prediction complete. Output saved as 'Testing Data with
Predictions.csv'.")
```

Figure 4: Python code used for k-NN classification

To prepare the data, the input features were extracted and standardized to a common scale to allow accurate distance-based comparisons between records. Standardization was essential due to the varying units and ranges of the input variables. The prepared data was then passed to a K-Nearest Neighbors regressor with k = 3, chosen to balance local prediction sensitivity and generalization. The K-NN model does not generate a traditional equation but instead makes predictions based on the average outcomes of the three most similar data points in the training set. This made the analysis highly dependent on the structure and distribution of the training data.

4.2 Decision Tree Analysis

Figure 5 shows the decision tree classification model developed using the CART algorithm in Minitab 22 to categorize pavement condition as either "Good" or "Unsatisfactory." The model used six predictors: ESALs, pavement thickness, temperature, precipitation, drainage presence, and rut depth. Table 2 shows the entropy calculated for all the variables before entering the dataset into Minitab for analysis.

Table 2: Entropy Values for all the Variables

Variables	ESALs (kESALs/yr)	Drainage Flag (0/1)	Thickness (mm)	Temperature (°C)	Precipitation (mm)	RUT (mm)
Good	0.70	0.65	0.24	0.15	0.57	0.52
Unsatisfactory	0.30	0.35	0.76	0.85	0.43	0.48
Entropy	0.85	0.88	0.73	0.88	0.87	0.80

The decision tree begins by splitting the data based on pavement thickness, which emerged as the most influential variable. Segments with a thickness less than or equal to 4.87 mm were classified entirely as "Good." For thicker pavements, the next important factor was rut depth, with a threshold of 3.5 mm helping to further separate the classes.

Additional splits were made using temperature and thickness again, showing that interactions between these variables played a role in determining pavement condition. For example, pavements with higher rut depths and temperatures above 7.65°C were more likely to be classified as "Unsatisfactory."

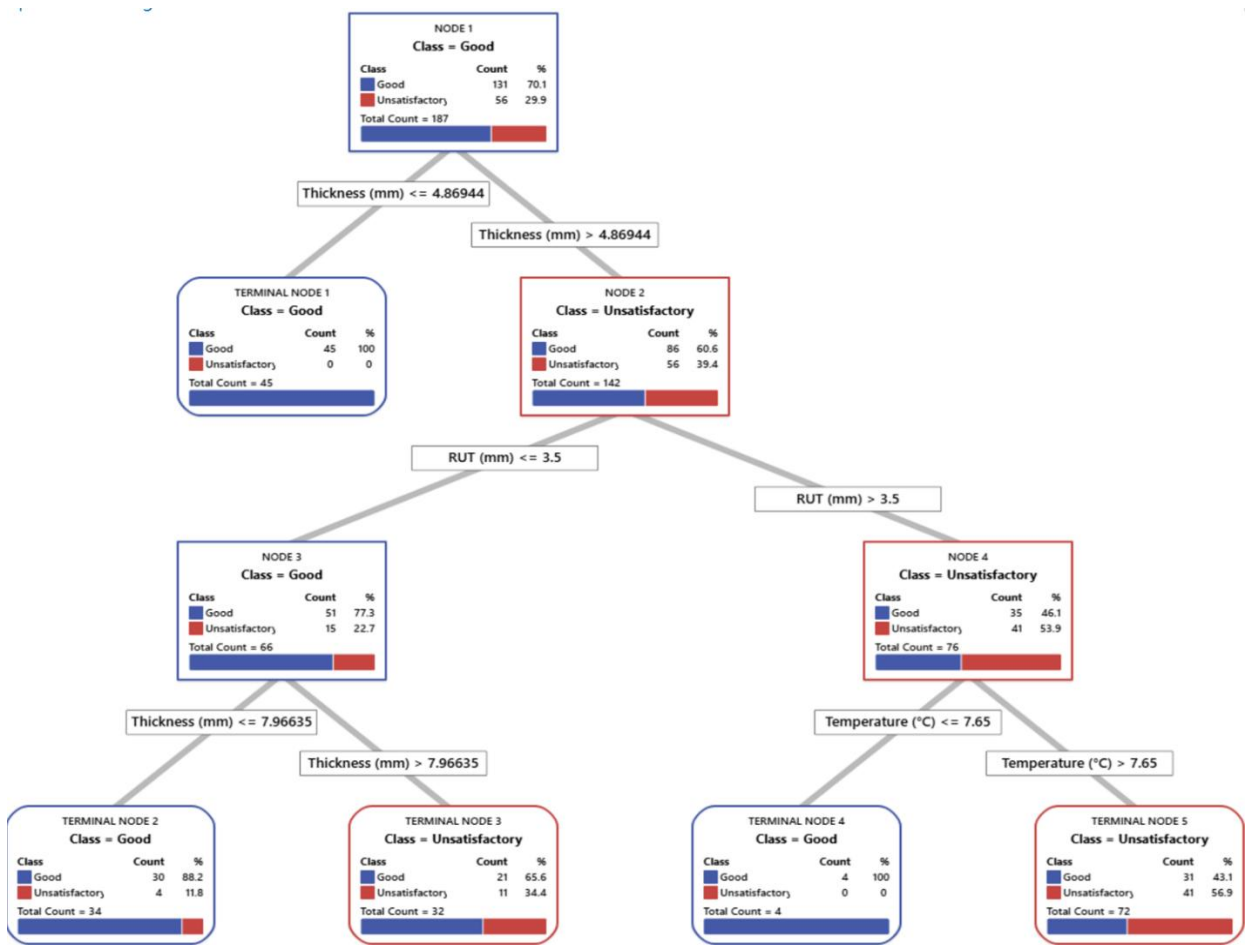

Figure 5: Decision tree diagram used to classify pavement condition

The tree structure revealed that thickness, rut depth, and temperature were the key drivers in the classification, with the model effectively identifying patterns that align with engineering understanding of pavement performance.

5. Results

5.1 kNN Results

The accuracy of the kNN model was evaluated using a separate dataset consisting of 45 pavement sections. A representative subset from the testing dataset, including the predicted and actual rut depth values, is shown in Table 3. The full testing dataset details are presented in Appendix B.

Table 3: Sample of Testing Dataset and Predictions (kNN)

STATE	ESALS (kESAL/yr)	Drainage Flag (0/1)	Thickness (mm)	Average Annual Temperature (°C)	Average Annual Precipitation (mm)	Predicted Rut(mm)	Actual Rut(mm)
Wisconsin	352	1	13.47	7.5	898.2	3	4
California	275	1	7.27	19.9	168.9	2.75	2
Colorado	515	1	6.87	11	523.3	5	5
Kansas	840	1	5.6	12.7	855.3	4.33	8

The predictive accuracy of the K-Nearest Neighbors (K-NN) model was assessed through a scatter plot comparing actual versus predicted rut depths, as illustrated in Figure 6. In this plot, each point corresponds to a pavement section from the test dataset, with the x-axis representing the observed rutting measurements and the y-axis depicting the model's predicted values.

The red dashed line signifies the ideal prediction scenario (y = x), where predictions perfectly match actual measurements. Points situated closer to this line indicate higher prediction accuracy. The plot reveals that the model's predictions closely align with actual values, particularly for rut depths ranging from 2 mm to 5 mm. However, there's a noticeable dispersion for higher rut depths, suggesting that while the model performs well for moderate rutting, its accuracy diminishes in extreme cases.

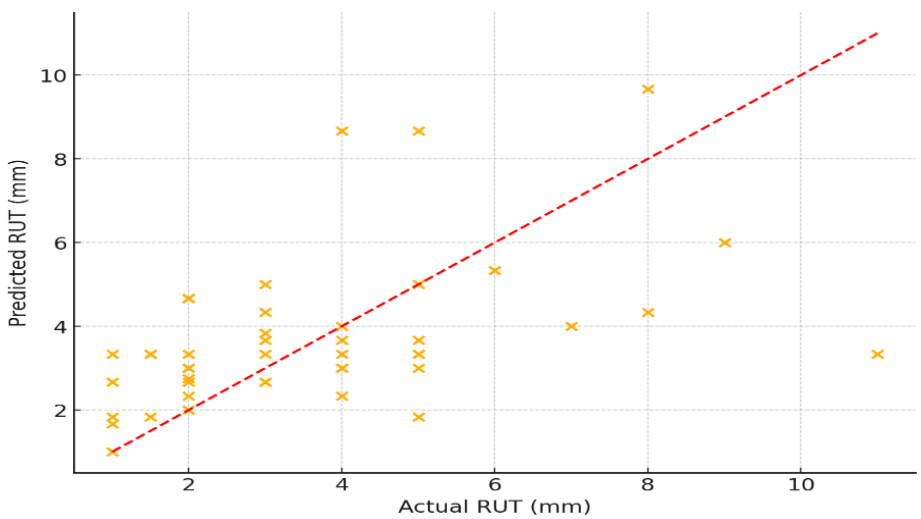

Figure 6: Actual Rut vs. Predicted Rut

The model's performance was quantitatively evaluated using the Sum of Squared Errors (SSE), which amounted to 205.03. This metric encapsulates the total squared differences between predicted and actual values, indicating a reasonable predictive capability of the model. Nonetheless, the observed variability points to potential areas for refinement to enhance accuracy, especially for extreme rutting scenarios.

5.2 Decision Tree Results

The accuracy of the Decision model was evaluated using a separate dataset consisting of 48 pavement sections. A representative subset from the testing dataset, including the predicted and

actual rut depth values, is shown in Table 4. The full testing dataset details are presented in Appendix C.

Table 4: Sample of Testing Dataset and Predictions (Decision Tree)

STATE	ESALS (kESAL/yr)	Drainage Flag (0/1)	Thickness (mm)	Average Annual Temperature (°C)	Average Annual Precipitation (mm)	RUT (mm)	Pavement Condition
Arizona	2354	0	8.1	23.3	190.6	5	Unsatisfactory
California	334	0	5.15	24	65.9	2	Unsatisfactory
Arizona	2351	0	8.5	23.3	190.6	4	Unsatisfactory
Washington	722	0	29.36	10	284.2	2	Good

The Decision Tree model was tested on a dataset comprising 48 pavement sections, aiming to categorize each as either "Good" or "Unsatisfactory." As depicted in Figure 7, the model accurately classified 20 sections as "Good" (true positives) and 19 as "Unsatisfactory" (true negatives). However, it misclassified 7 "Unsatisfactory" sections as "Good" (false positives) and 2 "Good" sections as "Unsatisfactory" (false negatives).

Reality \ Prediction	Good	Unsatisfactory
Good	20	7
Unsatisfactory	2	19

Figure 7: Testing Efficiency (the rows representing reality vs. the columns representing prediction)

These outcomes resulted in an overall accuracy of 81.25%, a precision of 74.07%, a recall of 90.91%, and an F1-score of 81.63%. The high recall indicates the model's effectiveness in correctly identifying pavements in good condition, which is crucial for maintenance planning. The balance between precision and recall suggests the model's reliability in practical pavement condition assessments.

6. Conclusion

This study applied two machine learning techniques to assess pavement performance using common variables: Equivalent Single Axle Loads (ESALs), pavement thickness, annual temperature, annual precipitation, and drainage presence.

1. A K-Nearest Neighbors (K-NN) regression model predicted rut depth, with a Sum of Squared Errors (SSE) of 205.03 on a test set of 45 pavement sections. It worked well for 2–5 mm rut depths but less accurately for higher values.

2. A Decision Tree classification model, using the CART algorithm, classified pavement condition as "Good" or "Unsatisfactory," achieving 81.25% accuracy, 74.07% precision,

90.91% recall, and an 81.63% F1-score on 48 test sections. Thickness, rut depth, and temperature were most influential.

3. The Decision Tree model's high recall value is beneficial for identifying good pavement for preventive maintenance.

Overall, both models show the potential of data-driven approaches in pavement management. Future work should expand the dataset, improve variable definitions (especially drainage), and test ensemble methods for better accuracy and generalization.

References

Alnaqbi, A. J., Zeiada, W., Al-Khateeb, G. G., Hamad, K., & Barakat, S. (2023). Creating rutting prediction models through machine learning techniques utilizing the Long-Term Pavement Performance database. *Sustainability, 15*(18), 13653. https://doi.org/10.3390/su151813653

Federal Highway Administration. (1993). *Sensitivity Analyses for Selected Pavement Distresses* (SHRP-P-392). Strategic Highway Research Program.

Federal Highway Administration. (2006). *Long-Term Pavement Performance Program: Pavement Performance in Frost Areas—Comprehensive Report* (FHWA-HRT-06-121). U.S. Department of Transportation.

Federal Highway Administration. (2017). *Distress Identification Manual for the Long-Term Pavement Performance Program* (5th Rev. ed.). U.S. Department of Transportation.

Finn, F. N., Saraf, C. L., Kulkarni, R. S., Nair, K. G., Smith, W., & Abdullah, A. (1976). Development of pavement structural subsystems. *Transportation Research Record, 616*, 9–19.

Ghanim, M. S., Abu-Samra, S., García, G. A. C., & Zayed, T. (2023). Explainable machine learning for predicting the International Roughness Index of flexible pavements. *Sustainability, 15*(12), 9617. https://doi.org/10.3390/su15129617

Gedafa, D. S., Hossain, M., Schieber, G. L., Van, V., & Esmaeili, A. H. (2021). Prediction of Pavement Condition Index and International Roughness Index using machine learning algorithms. *Journal of Infrastructure Systems, 27*(3), 04021018. https://doi.org/10.1061/(ASCE)IS.1943-555X.0000602

Jaafar, Z. F., & Uddin, W. (2015). Modeling of pavement rutting deterioration for LTPP Southern Region using multiple linear regression. In *Proceedings of the 6th International Conference on Bituminous Mixtures and Pavements* (pp. 713–722). Thessaloniki, Greece.

Lajnef, S. B., Raddaoui, M., & Perraton, D. (2020). Developing a prediction model for rutting depth of asphalt mixtures using gene expression programming. *Materials, 13*(16), 3598. https://doi.org/10.3390/ma13163598

Mazari, M., Rodriguez, D., & El-Gendy, A. (2021). Climate change impact on infrastructure: A machine learning solution for predicting pavement condition index. *Cleaner Engineering and Technology, 4*, 100231. https://doi.org/10.1016/j.clet.2021.100231

Sousa, H., Almeida, J. C., Couto, A., & Bastos, J. (2023). Machine learning approaches for International Roughness Index prediction on flexible pavements: A systematic review. *Applied Sciences, 13*(24), 13254.

Appendix A – Training Dataset

STATE	ESALS (kESAL/yr)	Drainage Flag (0/1)	Thickness (mm)	Temperature (°C)	Precipitation (mm)	RUT (mm)	Pavement Condition
Arizona	541	0	6.9	23.3	185.3	6	Unsatisfactory
California	1291	0	7.2	17.7	179.2	9	Unsatisfactory
Delaware	221	0	16.2	14.1	1089.2	2	Good
Indiana	572	1	2.6	13.5	1563.3	8	Good
Iowa	411	0	12.9	9.6	1107.7	2	Good
Kansas	917	0	6.6	12.9	795.9	5	Unsatisfactory
Louisiana	75	0	5.6	19.1	1544.4	6	Unsatisfactory
Michigan	233	0	3.1	9.9	867.2	4	Good
Missouri	259	0	2.5	15.5	1122.3	1	Good
Montana	206	0	5.4	8.7	387.9	16	Unsatisfactory
Nevada	184	0	5.4	11.0	265.9	1.5	Good
New Mexico	966	0	5.5	13.2	394.7	2	Good
Ohio	398	0	5.5	12.2	931.3	2	Good
Oklahoma	50	0	6.2	17.8	886.4	3	Good
Oregon	161	0	9.2	10.2	288.4	3	Good
Pennsylvania	132	1	6.4	9.3	1365.3	2	Good
Texas	107	0	3.9	18.9	889.9	4	Good
Vermont	25	1	10.1	9.2	677.4	2	Good
Virginia	214	0	5.9	14.8	1757.0	1	Good
Wisconsin	2	0	13.5	6.0	853.4	2	Good
Manitoba	121	1	3.1	4.0	461.5	2	Good
Delaware	227	0	9.8	14.1	1089.2	1	Unsatisfactory
Kansas	768	0	6.6	13.2	902.6	1	Unsatisfactory
Colorado	530	1	6.9	11.0	265.0	7	Unsatisfactory
Nevada	232	0	5.7	11.4	98.5	2	Good
Arizona	643	1	6.2	23.6	188.4	6	Unsatisfactory
Montana	197	0	4.8	8.7	387.9	8	Good
Oregon	154	0	3.1	9.8	359.4	5	Good
Washington	125	0	11.8	10.0	502.8	12	Good
Arizona	706	0	7.1	23.5	216.0	3	Good
California	1291	0	7.1	17.7	179.2	4	Good
Wisconsin	401	0	13.9	7.8	965.6	4	Unsatisfactory
Ohio	701	1	6.6	9.6	924.3	2	Good
Oklahoma	49	1	8.8	17.8	886.4	2	Good
Washington	131	0	4.5	13.4	188.0	4	Good
Arizona	1635	1	6.5	23.3	245.9	10	Unsatisfactory
Delaware	223	0	19.0	14.1	1089.2	5	Good
Iowa	412	1	10.8	9.6	1107.7	2	Good
Arizona	2336	0	7.3	23.3	190.6	2	Good
Washington	795	0	19.2	9.2	285.4	13	Good
Colorado	542	0	11.9	11.5	376.0	4	Good
Colorado	506	1	5.7	11.0	523.3	3	Good
California	256	0	4.9	18.9	357.6	2	Unsatisfactory
Arizona	541	1	5.5	23.3	185.3	3	Unsatisfactory
Washington	661	0	42.6	10.1	297.7	7	Unsatisfactory
Washington	699	0	42.6	10.0	284.2	7	Unsatisfactory
Oklahoma	49	0	11.3	17.8	886.4	3	Good
Iowa	412	0	13.8	9.6	1107.7	2	Unsatisfactory

Wisconsin	2	0	12.8	6.0	853.4	3	Good
Missouri	225	0	2.5	15.0	1034.1	2	Good
Colorado	1060	0	8.9	11.0	265.0	5	Unsatisfactory
Arizona	42	1	6.6	23.6	193.9	4	Unsatisfactory
Arizona	639	1	5.6	23.6	188.4	6	Good
Arizona	710	0	8.1	23.5	216.0	4	Unsatisfactory
Missouri	259	0	2.5	14.2	1007.9	3	Good
Washington	722	0	23.8	10.0	284.2	3	Unsatisfactory
Arizona	2339	1	5.6	23.3	190.6	3	Good
Arizona	2618	0	6.2	23.3	245.9	6	Good
New Mexico	967	0	5.4	13.2	394.7	1	Good
Virginia	213	1	4.7	14.8	1757.0	2	Good
Florida	19	1	8.2	22.0	1801.7	4	Unsatisfactory
Oregon	160	0	9.2	10.0	298.1	2	Good
Colorado	454	0	7.2	10.4	307.2	7	Good
Arizona	1482	0	1.9	23.6	103.9	2.5	Good
Washington	348	0	23.8	10.3	411.0	4	Unsatisfactory
Oklahoma	38	1	3.4	17.2	625.6	1.5	Good
Arizona	710	0	8.8	23.5	216.0	3	Unsatisfactory
Arizona	544	0	8.8	23.3	185.3	5	Unsatisfactory
Missouri	259	0	2.6	15.5	1122.3	1	Good
Washington	823	0	23.8	9.2	285.4	5	Unsatisfactory
Missouri	259	0	2.5	14.4	1104.8	3	Good
Colorado	503	0	8.9	11.1	307.5	6	Unsatisfactory
Nevada	349	0	6.9	11.8	269.4	3	Good
Louisiana	72	0	5.3	19.1	1544.4	8	Good
Washington	803	1	16.7	9.2	285.4	9	Unsatisfactory
Vermont	272	1	11.0	8.4	907.9	3	Good
California	96	0	6.1	14.3	1054.0	3	Good
Arizona	710	1	6.2	23.5	216.0	3	Good
Nevada	273	0	7.5	11.2	182.2	3	Good
Colorado	503	0	11.9	11.1	307.5	5	Good
Kansas	768	0	6.5	13.2	902.6	3	Good
Pennsylvania	2090	1	9.7	11.4	1870.1	8	Good
Virginia	211	1	4.0	14.8	1757.0	1.5	Good
New Mexico	921	0	5.4	13.1	246.9	3	Good
Wisconsin	2	1	5.3	6.0	853.4	3	Unsatisfactory
Missouri	259	0	2.5	14.4	1104.8	4	Good
Wisconsin	2	0	9.2	6.0	853.4	2	Unsatisfactory
Missouri	259	0	2.5	14.4	1104.8	2	Good
Washington	2	0	34.6	13.1	328.2	2	Good
Washington	58	0	11.8	10.8	375.5	11	Good
Arizona	709	0	8.5	23.5	216.0	3	Good
Washington	823	0	29.4	9.2	285.4	4	Good
Colorado	530	0	11.9	11.0	265.0	6	Good
California	447	1	6.8	19.0	206.1	4.25	Good
Iowa	412	0	14.0	9.6	1107.7	3	Unsatisfactory
Arizona	667	0	3.4	13.4	163.2	3	Good
Washington	94	0	4.5	8.6	636.3	1	Good
Colorado	459	1	6.9	10.4	307.2	7	Unsatisfactory
Colorado	452	0	7.7	10.2	422.0	13	Unsatisfactory
Ohio	401	1	5.3	12.3	944.4	4	Good
Wisconsin	2	1	6.6	6.0	853.4	1	Good
Delaware	228	1	12.3	14.1	1089.2	3	Good
Manitoba	120	1	2.9	4.0	461.5	2	Good

Wisconsin	293	0	11.8	7.1	958.6	1	Good
Wisconsin	344	0	6.8	7.5	898.2	4	Good
Washington	563	0	7.9	13.4	161.0	7	Good
Colorado	514	0	8.7	11.0	523.3	6	Unsatisfactory
Arizona	2335	1	5.5	23.3	190.6	6	Good
Arizona	2338	0	6.2	23.3	190.6	6	Good
Washington	348	1	16.7	10.3	411.0	7	Unsatisfactory
Missouri	225	0	2.5	15.0	1034.1	2	Good
Arizona	42	1	5.6	23.6	193.9	3	Good
Florida	0	0	2.7	20.3	1443.0	5	Good
Montana	200	1	5.9	8.7	387.9	14	Unsatisfactory
California	606	0	1.8	24.1	123.3	3	Good
California	178	0	5.2	24.1	54.6	3	Good
Michigan	231	0	3.4	9.9	867.2	2	Good
Colorado	565	0	7.3	11.0	253.5	4	Good
Montana	191	0	5.6	7.2	482.8	6	Good
California	286	1	7.2	20.4	310.8	2	Good
Colorado	576	1	6.4	11.0	253.5	4	Unsatisfactory
Arizona	437	0	3.4	12.7	339.0	3	Good
Oklahoma	49	0	31.7	17.8	886.4	1	Good
Manitoba	246	1	2.6	3.6	338.6	1	Good
Arizona	544	1	6.3	23.3	185.3	2	Good
California	287	1	6.8	20.3	306.2	2	Good
Arizona	573	0	2.1	23.7	85.0	4	Good
Arizona	1447	0	1.7	23.6	87.7	2	Good
California	289	1	7.3	19.7	323.8	2	Good
Kansas	934	0	7.8	12.9	795.9	4	Unsatisfactory
Wisconsin	2	1	13.5	6.0	853.4	4	Good
Iowa	412	0	17.5	9.6	1107.7	5	Unsatisfactory
Colorado	530	1	6.4	11.0	265.0	5	Unsatisfactory
Ohio	908	1	8.4	12.2	933.6	3	Unsatisfactory
Colorado	459	0	11.9	10.2	422.0	4	Good
California	339	0	31.9	15.5	445.2	1	Good
Washington	157	0	4.4	9.3	460.7	2	Good
Kansas	923	1	5.6	12.9	795.9	8	Good
Colorado	898	0	6.9	10.2	422.0	10	Unsatisfactory
Texas	103	0	3.9	18.5	619.2	5	Good
Washington	180	0	4.4	9.8	570.7	1	Good
Oklahoma	49	1	15.1	17.8	886.4	2	Good
Kansas	1675	0	6.7	12.7	855.3	5	Unsatisfactory
Oklahoma	38	1	3.5	16.2	746.6	2	Good
Vermont	168	1	9.3	8.4	910.0	3	Good
Florida	0	0	2.8	21.1	1699.7	3	Good
Arizona	1055	1	6.3	23.7	99.3	1	Good
California	1298	0	8.7	17.7	179.2	3	Unsatisfactory
Washington	700	1	16.7	10.5	194.3	9	Unsatisfactory
Delaware	228	1	12.5	14.1	1089.2	1	Unsatisfactory
Iowa	412	1	10.1	9.6	1107.7	5	Unsatisfactory
Washington	103	0	11.8	9.7	405.8	9	Good
Ohio	1325	1	6.5	12.1	1069.4	3	Good
Arizona	706	1	5.6	23.5	216.0	4	Good
Wisconsin	388	0	7.0	7.8	965.6	6	Unsatisfactory
Wisconsin	353	0	13.8	7.5	898.2	2	Unsatisfactory
Oklahoma	96	0	3.6	17.8	891.2	1	Good
Washington	130	0	4.5	8.6	511.6	4	Good

Montana	384	1	6.3	8.7	387.9	4	Good
Wisconsin	346	0	7.0	7.5	898.2	4	Good
Washington	659	0	19.2	10.1	297.7	10	Good
Arizona	643	0	8.7	23.6	188.4	4	Unsatisfactory
Oregon	154	0	3.1	9.8	359.4	5	Good
Washington	660	0	35.1	10.1	297.7	4	Unsatisfactory
Virginia	195	0	4.0	14.1	1098.4	1	Good
Colorado	454	1	5.7	10.4	307.2	6	Good
Manitoba	120	1	2.6	4.0	461.5	2	Good
Washington	157	0	4.7	8.6	636.3	1	Good
Arizona	84	0	7.1	23.6	193.9	4	Good
Arizona	2335	0	7.1	23.3	190.6	4	Good
Wisconsin	2	1	10.4	6.0	853.4	2	Unsatisfactory
Michigan	233	1	2.4	9.9	867.2	2	Good
Nevada	184	0	5.7	11.0	265.9	1	Good
Wisconsin	286	1	10.4	7.1	958.6	2	Good
Arizona	2648	1	6.6	23.3	245.9	4	Unsatisfactory
Arizona	643	1	6.3	23.6	188.4	4	Unsatisfactory
Ohio	69	0	3.9	11.1	1037.8	3	Good
Texas	128	0	4.0	17.9	861.9	9	Good
Arizona	2644	1	6.3	23.3	245.9	3	Good
Colorado	459	1	6.9	10.2	422.0	5	Unsatisfactory
Ohio	700	1	9.0	9.6	924.3	2	Good
Colorado	564	0	7.2	11.0	253.5	5	Good
Missouri	259	0	2.5	14.2	1007.9	5	Good
Kansas	935	0	7.9	12.9	795.9	6	Unsatisfactory
Louisiana	73	1	4.6	19.1	1544.4	3	Good
Arizona	1411	0	6.9	23.5	216.0	6	Unsatisfactory
Arizona	2647	0	8.7	23.3	245.9	4	Good

Appendix B – Testing Dataset for kNN

STATE	ESALS (kESAL/yr)	Drainage Flag (0/1)	Thickness (mm)	Temperature (°C)	Precipitation (mm)	RUT (mm)	Predicted_RUT (mm)
Wisconsin	352	1	13.46667	7.5	898.2	4	3
California	275	1	7.266667	19.9	168.9	2	2.75
Colorado	515	1	6.866667	11	523.3	5	5
Kansas	840	1	5.6	12.7	855.3	8	4.333333
Washington	149	0	3.238462	8.6	636.3	1	1
Arizona	545	1	6.566667	23.3	185.3	5	3
Missouri	260	0	2.509091	14.4	1104.8	2	3
Oregon	192	0	9.207692	10.7	236.8	3	2.666667
Colorado	533	0	7.25	11.5	376	5	8.666667
Arizona	638	1	5.5	23.6	188.4	6	5.333333
Oklahoma	37	1	3.469231	16	908.4	5	1.833333
Arizona	545	0	8.75	23.3	185.3	3	5
Delaware	221	1	10.05	14.1	1089.2	2	3
Oklahoma	49	0	15.18421	17.8	886.4	2	3.333333
Oklahoma	96	0	3.570588	17.8	891.2	2	4.666667
Wisconsin	344	1	10.425	7.5	898.2	2	2.666667
Virginia	192	0	5	14.1	1098.4	1	2.666667
Arizona	527	0	3.4	13.1	261.1	4	3
Delaware	227	0	10.73333	14.1	1089.2	2	2
Oregon	161	0	3.053846	10.2	288.4	3	3.833333
Ohio	909	1	9.025	12.2	933.6	3	4.333333
Manitoba	255	1	3.128571	3.6	338.6	1	1.666667
Washington	722	0	15.6	10	284.2	8	9.666667
Arizona	709	1	6.266667	23.5	216	3	3.666667
Arizona	2647	1	6.233333	23.3	245.9	4	3.333333
Washington	715	0	29.36	10.5	194.3	4	3.666667
Oregon	158	0	9.253846	10	298.1	2	4.666667
Oklahoma	49	0	11.632	17.8	886.4	2	2.333333
Wisconsin	287	0	6.875	7.1	958.6	2	4.666667
Arizona	706	0	7.2	23.5	216	3	3.333333
Arizona	2647	0	7.6	23.3	245.9	7	4
California	234	0	3.850877	24.3	57.6	5	3.666667
Missouri	257	0	2.536364	14.1	1002.8	1.5	3.333333
Wisconsin	2	0	6.95	6	853.4	5	3.333333
Nevada	184	0	6.9	11	265.9	1.5	1.833333
California	2580	0	6.95	17.7	179.2	4	4
Wisconsin	2	0	14	6	853.4	4	2.333333
Missouri	257	0	2.545455	14.1	1002.8	1	3.333333
Colorado	505	0	7.15	11	523.3	4	8.666667
Washington	1097	0	7.9	12.1	211.5	11	3.333333
Washington	673	0	37.025	10.1	297.7	9	6
Missouri	258	0	2.509091	14.1	1002.8	1.5	3.333333
Oklahoma	49	1	3.384615	16.6	1004.2	1	1.833333
Arizona	710	0	7.6	23.5	216	5	3.333333
Arizona	1056	1	6.233333	23.7	99.3	3	2.666667

Appendix C – Testing Dataset for Decision Tree

STATE	ESALS (kESAL/yr)	Drainage Flag (0/1)	Thickness (mm)	Temperature (Â°C)	Precipitation (mm)	RUT (mm)	IRI (m/km)	Pavement Condition
Arizona	2354	0	8.1	23.3	190.6	5	1.6686	Unsatisfactory
California	334	0	5.158974	24	65.9	2	1.6314	Unsatisfactory
Arizona	2351	0	8.5	23.3	190.6	4	1.5084	Unsatisfactory
Washington	722	0	29.36	10	284.2	2	1.2282	Good
California	1504	0	5.1	18.9	370.2	2	1.280333	Good
Arizona	1415	0	1.872222	23.6	87.7	2	0.794	Good
Arizona	531	0	3.257143	13.1	261.1	4	0.815	Good
Colorado	502	0	8.7	11.1	307.5	7	1.9774	Unsatisfactory
Arizona	2647	0	8.75	23.3	245.9	2	1.538053	Unsatisfactory
California	2580	0	7	17.7	179.2	6	1.716	Unsatisfactory
Ohio	701	0	8.65	9.6	924.3	2	1.155	Good
Washington	180	0	4.372727	9.8	570.7	1	0.7675	Good
Montana	196	0	4.838889	7.2	482.8	7	0.7122	Good
Oklahoma	49	1	3.423077	16.6	1004.2	1	0.8516	Good
Colorado	528	0	6.9	11.5	376	9	1.878	Unsatisfactory
Arizona	706	0	7.25	23.5	216	2	1.2126	Good
Washington	677	0	29.36	10.1	297.7	2	1.201	Good
Arizona	1040	0	6.85	23.7	99.3	5	2.7798	Unsatisfactory
Oklahoma	96	0	3.747059	17.8	891.2	2	0.5778	Good
Kansas	779	0	6.225	13.2	902.6	3	1.7028	Unsatisfactory
Arizona	2610	0	7.1	23.3	245.9	7	3.502474	Unsatisfactory
Arizona	42	0	8.65	23.6	193.9	5	1.7996	Unsatisfactory
Missouri	257	0	2.509091	14.1	1002.8	1	0.5758	Good
Washington	676	0	15.6	10.1	297.7	8	1.5126	Unsatisfactory
Colorado	452	0	7.25	10.2	422	5	1.2258	Good
Ohio	401	1	5.352381	12.3	944.4	1	0.8614	Good
Missouri	259	0	2.545455	15.5	1122.3	1	0.6534	Good
New Mexico	1239	0	5.266667	12.5	337.7	2	0.6734	Good
Louisiana	72	0	5.581818	19.1	1544.4	6	0.7502	Good
California	1299	1	6.433333	17.7	179.2	5	1.8748	Unsatisfactory
Kansas	917	0	6.666667	12.9	795.9	6	2.5016	Unsatisfactory
Arizona	1056	0	8.8	23.7	99.3	5	2.6096	Unsatisfactory
Arizona	2335	0	7.1	23.3	190.6	7	3.6342	Unsatisfactory
Oklahoma	37	1	3.492308	16	908.4	4	0.9014	Good
Michigan	229	1	3.088372	9.9	867.2	2	0.5664	Good
Ohio	429	1	6.16	10.6	963.5	2	0.8238	Good
Arizona	601	0	1.688889	23.7	85	5	0.8902	Good
Colorado	456	0	8	10.2	422	6	1.837	Unsatisfactory
Colorado	559	0	6.9	11	253.5	7	2.1542	Unsatisfactory
Washington	348	0	35.08	10.3	411	3	1.9029	Unsatisfactory
Pennsylvania	4028	1	22.1	10.1	1189.8	3	1.0798	Good
Washington	710	0	37.025	10.5	194.3	9	1.388	Good
Washington	80	0	4.481818	9.3	460.7	1.5	0.794	Good
Virginia	192	0	5.930769	14.1	1098.4	1	1.3564	Good
New Mexico	917	0	5.488889	13.1	246.9	3	0.6716	Good
Arizona	643	0	8.75	23.6	188.4	3	1.533	Unsatisfactory
Kansas	686	0	6.225	13.4	805.3	7	1.7442	Unsatisfactory
Arizona	441	0	3.257143	12.7	339	3	0.9058	Good